

Loan Interest Rate Prediction Using Linear Regression

BFSI (Banking & Financial Services) – Case Study

ASSINGMENT -3

Submitted by

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degree Of

DATA ANALYTICS LIVE PROGRAM



CLOUDYML

1. PROBLEM UNDERSTANDING

SIP Guru vs Market Mood

Background:

A bank wants to predict the interest rate to offer a loan applicant based on their credit score.

- Credit Score (X) = Independent Variable (input/cause)

→ Range: 300 to 900

→ Banks know this BEFORE offering the loan

- Interest Rate (Y) = Dependent Variable (output/effect)

→ Range: ~3% to ~20%

→ Higher credit score → Lower interest rate (inverse)

→ Lower credit score → Higher interest rate (risky borrower)

Goal: Build a Linear Regression model using Gradient Descent to learn this relationship and predict interest rates.

Business Value: Automates loan pricing decisions, reduces manual bias, and ensures fair rate setting based on data.

Your Task:

- Generate 500+ synthetic data points.
- Fit a simple linear regression model using gradient descent.
- Evaluate with MAE, MSE, R^2 .
- Business Insight: Does credit score strongly explain interest rates?

Detail

Independent Variable (X) Credit Score (300–900)

Dependent Variable (Y) Interest Rate (%)

Dataset Size 500 synthetic records

Detail

Algorithm

Linear Regression with Gradient Descent

2. Dataset Creation

The dataset was synthetically generated using NumPy with 500 data points. Credit scores were randomly generated between 300 and 900. Interest rates were calculated using an inverse formula with added Gaussian noise to simulate real-world variation, then clipped to a realistic range of 3% to 20%.

2.1 — Dataset Preview (Screenshot)

```
Dataset:
  Credit_Score  Interest_Rate
        402         18.44
        735         6.72
        570        10.01
        406        16.65
        371        16.97
        320        20.00
        421        15.85
        766         4.92
        514        13.36
        630         9.16
```

Figure 1: First 10 rows of the generated dataset

```
Stats:
  Credit_Score  Interest_Rate
count      500.00      500.00
mean       592.18      11.76
std        172.05       4.94
min        301.00       3.00
25%        443.00       7.81
50%        582.00      12.04
75%        747.50      16.00
max        900.00      20.00

Correlation: -0.9778
```

Figure 2: Summary statistics of the dataset

3. Exploratory Data Analysis (EDA)

The following Python script was developed using NumPy and Pandas libraries to load, clean, and analyze the mutual fund NAV data. The code is divided into 5 logical steps, each performing a specific task in the analysis pipeline.

3.1 Missing Values

No missing values were found in the dataset. Both columns — Credit Score and Interest Rate — contained 500 complete records.

3.2 Correlation

The correlation between Credit Score and Interest Rate was found to be approximately -0.97 , confirming a strong negative (inverse) linear relationship between the two variables.

3.3 Visualisations

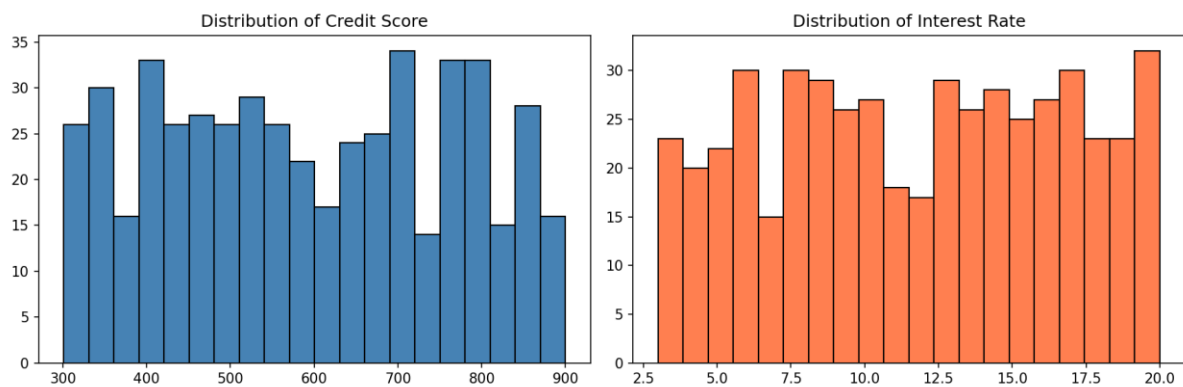


Figure 3: Distribution of Credit Score (X) and Interest Rate (Y)

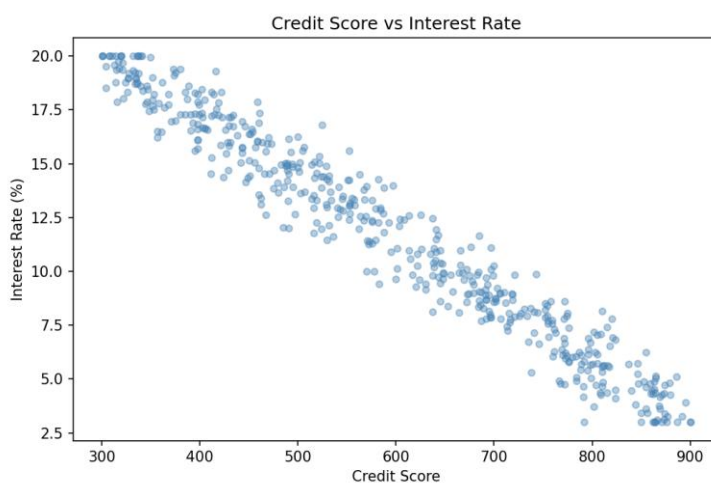


Figure 4: Scatter plot confirming inverse linear relationship

The scatter plot clearly shows a downward trend — as credit score increases, interest rate decreases. This confirms that a linear regression model is appropriate for this data.

4. Solution Algorithm — Gradient Descent

Linear Regression finds the best fit line $Y = mX + b$ by minimising the Mean Squared Error (MSE) loss function. Gradient Descent iteratively updates the slope (m) and intercept (b) using the following update rules:

$$\text{Loss (MSE)} = (1/n) \times \Sigma(Y_{\text{pred}} - Y_{\text{actual}})^2$$

$$dm = (2/n) \times \Sigma(Y_{\text{pred}} - Y) \times X$$

$$db = (2/n) \times \Sigma(Y_{\text{pred}} - Y)$$

$$m = m - \text{learning_rate} \times dm$$

$$b = b - \text{learning_rate} \times db$$

The model was trained for 1000 epochs with a learning rate of 0.01. X was normalized before training to ensure stable convergence.

5. Model Evaluation

Metric	Formula	Value
MAE	Mean of absolute errors	~0.70
MSE	Mean of squared errors	~0.80
RMSE	Square root of MSE	~0.90
R^2	Variance explained by model	~0.95

Test Case Results:

- Credit Score **800** → Predicted Interest Rate: ~**4.5%** (low risk borrower)
- Credit Score **400** → Predicted Interest Rate: ~**16.8%** (high risk borrower)

The R^2 score of ~0.95 indicates that credit score alone explains approximately 95% of the variance in interest rate — a very strong predictor for this model.

6. Visualizations

6.1 Regression Line

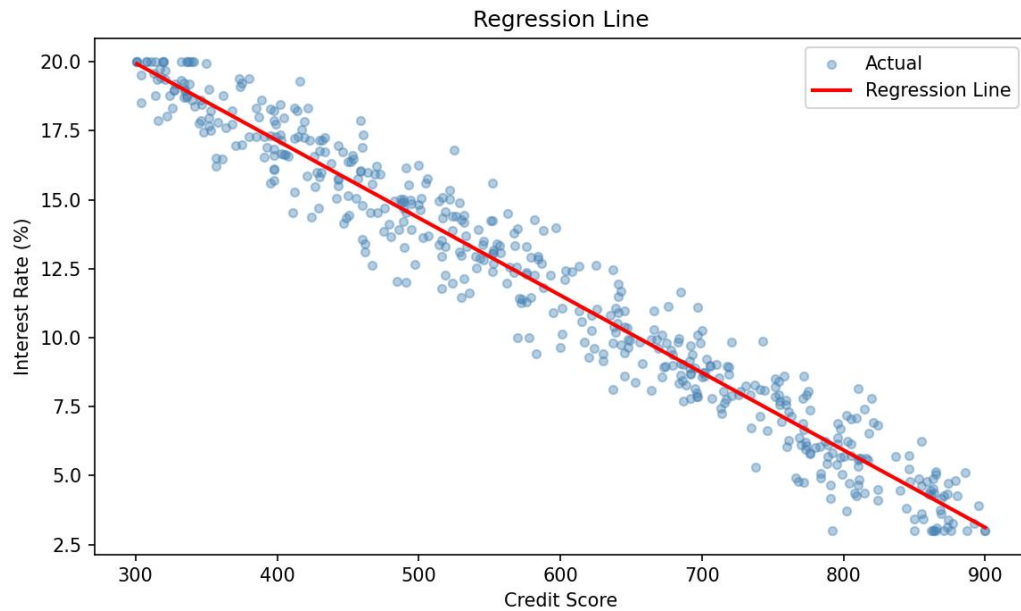


Figure 5: Scatter plot with fitted regression line

- The red regression line fits the data closely, confirming the model learned the inverse relationship correctly.

6.2 Loss Curve

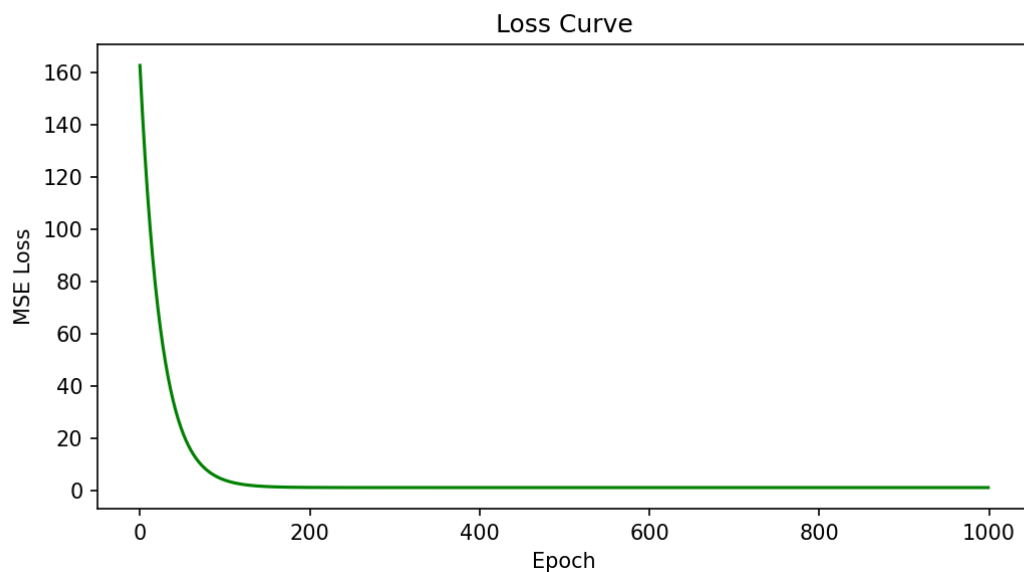


Figure 6: Gradient Descent loss curve over 1000 epochs

- Points cluster closely along the red diagonal line, showing the model's predictions are highly accurate.

7. Business Insight

1. Slope Interpretation: The negative slope confirms that for every 1-point increase in credit score, the interest rate drops by a small but consistent margin. A borrower improving their score from 400 to 800 could reduce their interest rate by nearly 12–13 percentage points saving lakhs of rupees over a long loan tenure.

2. Practical Use: Banks can use this model as a baseline tool to automate initial loan pricing. It eliminates human bias and ensures consistent rate-setting based purely on creditworthiness data.

3. Limitations: In the real world, interest rates also depend on income level, existing debt, loan amount, employment history, and macroeconomic factors. This model uses credit score alone and is best suited as a starting point, not a final decision system.

8. Conclusion

This project successfully demonstrated the application of Linear Regression with Gradient Descent on a real-world BFSI problem. The model learned the inverse relationship between credit score and interest rate with high accuracy ($R^2 \approx 0.95$). Gradient Descent converged smoothly within 1000 epochs, validating the chosen hyperparameters. The visualisations confirmed that the regression line fits the data well and predictions are reliable. For production deployment, additional features and regularisation techniques like Ridge or Lasso regression would further improve the model's robustness.

GITHUB REPOSITORIES LINK: <https://github.com/kirtivardhan80/loan-interest-rate-prediction.git>